

AI Internship – Day 4 Report

Understanding and Tuning Tree-Based Models (Decision Trees & Random Forest)

Name: Shafiq Ur Rehman

Department: Computer Science

Date: 5/6/2026

Objective

The objective of Day 4 was to:

- Understand tree-based machine learning models
 - Train Decision Tree classifiers with different depths
 - Analyze overfitting and underfitting behavior
 - Train Random Forest classifier with multiple trees
 - Evaluate model performance using accuracy
 - Interpret feature importance
 - Visualize decision tree behavior
-

Dataset Used

- **Dataset:** Titanic Dataset (Kaggle)
 - **Target Variable:** Survived (0/1)
 - **Features:** Passenger demographic and travel information
-

Concepts Learned

1. Decision Tree Classifier

A Decision Tree splits data into branches based on feature conditions to make predictions.

Example structure:

- If Sex = Female → Survived
 - If Sex = Male → Check Pclass / Age
-

2. Hyperparameters

✓ max_depth

Controls how deep the tree grows.

- Low depth → simple model (may underfit)
- High depth → complex model (may overfit)

✓ min_samples_leaf

Minimum samples required in a leaf node to prevent overfitting.

3. Overfitting vs Underfitting

Condition	Behavior
Underfitting	Model too simple, low accuracy
Overfitting	Model memorizes training data
Balanced	Good generalization

4. Model Training Results

Decision Tree (max_depth = 3)

```
Decision Tree (max_depth=3)
Train Accuracy: 0.8342696629213483
Test Accuracy: 0.7821229050279329
```

- Train Accuracy: **0.834**
- Test Accuracy: **0.782**

- ✓ Balanced model
 - ✓ Good generalization
-

Decision Tree (max_depth = 10)

```
Decision Tree (max_depth=10)
Train Accuracy: 0.9536516853932584
Test Accuracy: 0.7653631284916201
```

- Train Accuracy: **0.953**
- Test Accuracy: **0.765**

⚠ Overfitting observed

- ✓ High training accuracy but reduced test performance
-

5. Random Forest Classifier

- Number of Trees: 100

Results:

```
Random Forest
Train Accuracy: 1.0
Test Accuracy: 0.8100558659217877
```

- Train Accuracy: **1.0**
- Test Accuracy: **0.810**

- ✓ Improved performance compared to single decision tree
 - ✓ Better generalization using ensemble learning
-

6. Feature Importance Analysis

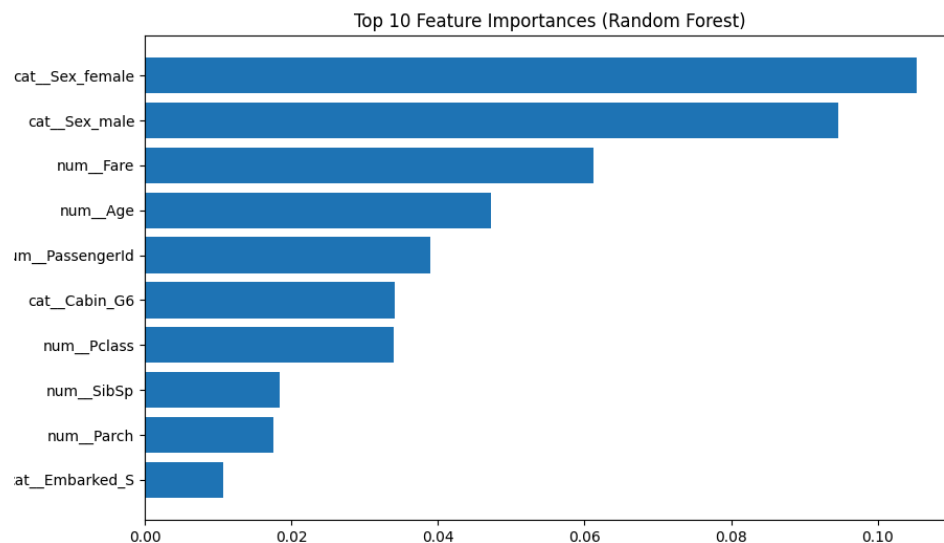
Random Forest was used to identify important features influencing survival.

Top influencing features:

- Sex
- Fare
- Pclass
- Age
- Embarked

7. Visualization

- Feature importance was plotted using matplotlib bar chart



These visualizations helped in understanding model decision-making.

Tools & Libraries Used

- Python
 - Pandas, NumPy
 - Scikit-learn:
 - DecisionTreeClassifier
 - RandomForestClassifier
 - Pipeline
 - ColumnTransformer
 - SimpleImputer
 - OneHotEncoder
 - Matplotlib
-

Key Learnings

- Decision Trees are highly interpretable models
 - Increasing depth increases risk of overfitting
 - Random Forest improves stability and accuracy
 - Feature importance helps interpret model behavior
 - Trade-off between bias and variance is crucial
 - Ensemble methods outperform single models in most cases
-

Conclusion

Day 4 focused on understanding tree-based machine learning models. Decision Trees were analyzed for different depths to observe overfitting and underfitting behavior. Random Forest was implemented to improve performance using ensemble learning. Feature importance analysis provided insights into the most influential variables affecting survival prediction. Overall, the models demonstrated improved understanding of interpretability and performance trade-offs in machine learning.